

ShopEZ – MERN Stack Project Documentation

1. Introduction

Project Title: ShopEZ – E-Commerce Application

Team Members: Add team member names and roles.

2. Project Overview

Purpose: ShopEZ is a MERN-based e-commerce platform that enables users to browse products, manage carts, place orders, and allows administrators to manage products and orders efficiently.

Features: User Authentication, Product Catalog, Cart Management, Order Processing, User Profiles, Admin Dashboard, Product Management, Order Tracking.

3. Architecture

Frontend: React.js with component-based architecture, routing, and API integration.

Backend: Node.js and Express.js implementing REST APIs using MVC architecture.

Database: MongoDB collections for Users, Products, Cart, Orders, and Admin.

4. Setup Instructions

Prerequisites: Node.js, npm, MongoDB, VS Code, Git, Postman.

Installation: Clone repository, install dependencies, configure environment variables, and start frontend/backend services.

5. Folder Structure

Client: src/components, src/pages, src/routes, src/assets

Server: controllers, models, routes, middleware, config

6. Running the Application

Frontend: npm run dev (client)

Backend: nodemon index.js (server)

7. API Documentation

Authentication APIs: Register, Login

Products APIs: CRUD operations

Cart APIs: Add, View, Remove Items

Orders APIs: Create and Manage Orders

8. Authentication

JWT-based authentication with password hashing and protected routes.

9. User Interface

Include screenshots of Landing Page, Products Page, Login/Register, Cart, Profile, Admin Dashboard, Orders, and Product Management.

10. Testing

Testing performed using Postman, integration testing, and manual functional testing.

11. Screenshots or Demo

Attach screenshots and demo video/GitHub repository links.

12. Known Issues

Limited payment integration, inventory synchronization, and responsive design improvements.

13. Future Enhancements

Wishlist, AI recommendations, payment gateways, notifications, real-time tracking, and multi-vendor support.

Conclusion

ShopEZ demonstrates a scalable MERN-stack e-commerce platform with efficient shopping and administrative workflows.